

Zeeshan Liaqat

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PROFILE

A results-driven Software Engineer with professional experience building mission-critical desktop and web applications across defence and commercial domains. Specializes in **WPF-based C2 and HMI systems**, **multi-sensor data fusion**, and **real-time hardware communication** (TCP/IP sockets, COM port/serial), combined with solid backend engineering using **ASP.NET Core** and **C#**. Experienced in designing operator-centric interfaces that consolidate complex, high-volume data streams into clean, actionable displays.

EXPERIENCE

Software Engineer

Shaheen Aero Traders

Nov. 2023 – Present

Rawalpindi, Pakistan

- Architected and developed a **Command & Control (C2) application** for an **Anti-Drone System** using **WPF** and **C#**, serving as the unified operator hub for commanding and monitoring interconnected **ESM (Electronic Support Measures)** and **ECM (Electronic Countermeasures)** subsystems.
- Engineered a **multi-sensor data fusion layer** that aggregates and correlates concurrent data streams from ESM, ECM, and radar subsystems, applying filtering and priority logic to surface only operationally relevant information and eliminate HMI clutter for field operators.
- Designed responsive **HMI screens** in WPF using the **MVVM** pattern, custom controls, and data binding to deliver real-time situational awareness dashboards — consolidating multi-source sensor feeds into a single, uncluttered operator view with context-sensitive alerts.
- Developed an **Anti-GNSS application** that communicates with RF amplifiers via **COM port (serial)** interfaces, implementing custom serial protocol handling for precise frequency, power, and device state control.
- Implemented **TCP/IP socket programming** within the Anti-GNSS application to exchange structured data with an external signal-processing system, managing connection lifecycle, message framing, and fault-tolerant reconnection logic.
- Applied **Asynchronous Programming** (async/await) and multithreading to manage concurrent hardware I/O streams — serial, socket, and sensor feeds — independently, keeping the UI fully responsive under continuous high-frequency data throughput.
- Architected and maintained scalable **ASP.NET Core** backend services to handle operational data logging, configuration management, and system health reporting, supporting both desktop C2 clients and web-based administrative interfaces.
- Developed and optimized **RESTful APIs** to facilitate seamless interoperability between WPF desktop clients, ASP.NET Core backend services, and hardware subsystems across the full system stack.
- Utilized **Entity Framework Core**, **LINQ**, and **SQL Server** for efficient persistence of mission logs, device configurations, sensor records, and inventory data, with optimized schemas and stored procedures for enterprise-grade reliability.
- Engineered modular, maintainable codebases following **SOLID principles** and design patterns (MVVM, Repository, Unit of Work) to ensure long-term testability and extensibility of mission-critical software.

EDUCATION

Comsats University Attock Campus

Bachelor of Science in Computer Science

Attock, Pakistan

Sep. 2019 – July 2023

Punjab College Jand, Attock

F.Sc Pre-Engineering (HSSC)

Attock, Pakistan

Aug. 2016 – June 2018

PROJECTS

Anti-Drone C2 System | WPF, C#, MVVM, Multi-Sensor Fusion, Real-Time HMI

Defence Project

- Built the **central C2 application** for an anti-drone platform, providing a unified operator interface to command and monitor **ESM** and **ECM** subsystems in real time.
- Implemented a **multi-sensor data fusion pipeline** that correlates concurrent feeds from ESM, ECM, and radar modules, applying priority and deduplication logic to reduce HMI noise and prevent operator overload.

- Designed **context-driven HMI screens** that dynamically surface only the most relevant information based on current threat state, minimizing cognitive load while maintaining full situational awareness.
- Handled edge cases including subsystem connection loss, hardware fault detection, and graceful recovery procedures to maintain **operational continuity** in live field deployments.

Anti-GNSS Control Application | *WPF, C#, Socket Programming, COM Port, Serial Protocols*

- Developed a **desktop control application** for an Anti-GNSS system managing RF amplifiers through **COM port (serial)** communication, with custom protocol framing for reliable command and response handling.
- Integrated **TCP/IP socket communication** to interface with an external signal-processing application, handling bidirectional message exchange, connection management, and automatic reconnection on link failure.
- Delivered a clean **WPF HMI** with real-time parameter monitoring and hardware configuration controls, enabling operators to manage complex RF hardware through a single, intuitive interface.

GradeWise.ai | *ASP.NET Core, React, SQL Server, RESTful APIs* | gradewise.ai

Web Project

- Developed the full-stack backend of a **GCSE & A-Level revision platform** serving **10,000+ students**, using ASP.NET Core Web API with real-time quiz sessions and progress tracking.
- Designed **SQL Server** schemas and Entity Framework Core data pipelines to manage a bank of 20,000+ questions, user performance records, and adaptive learning state.
- Built **RESTful APIs** consumed by a React frontend to power quiz delivery, student dashboards, and personalized feedback loops.

A.T.L.A.S. Engine | *ASP.NET Core, RESTful APIs, Multi-Channel Automation* | atlase.ai

Web Project

- Co-developed a **B2B lead generation platform** with an ASP.NET Core backend orchestrating a multi-stage pipeline: prospect discovery, contact enrichment, and automated multi-channel outreach.
- Designed and built **RESTful API services** supporting high-throughput data processing across enrichment, scheduling, and reporting workflows, achieving over **2,400 leads/month** in production.

TECHNICAL SKILLS

Languages: C#, Python, SQL, HTML5/CSS3

Desktop & HMI: WPF, MVVM, Custom Controls, Real-Time Data Binding, Multi-Sensor Data Fusion, C2 Systems

Communication & Hardware: TCP/IP Socket Programming, COM Port (Serial/RS-232), Hardware Protocol Integration

Web & Backend: ASP.NET Core, ASP.NET MVC, Web API, Entity Framework Core, RESTful APIs

Data: SQL Server, LINQ, Stored Procedures, Schema Design

Developer Tools: Visual Studio, VS Code, Git/GitHub, SSMS, Postman

Core Concepts: Multithreading, Async Programming, SOLID Principles, MVVM, Repository Pattern, OOP